

Arnab Mandal

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EDUCATION

Shiv Nadar University

B.Tech in Computer Science Engineering (8.59/10.0)

Noida, Delhi-NCR, India

Aug 2023 - Jul 2027

Heera Lal Public School, Class XII

Percentage : 91%

Madan Pur Dabas, Delhi-NCR, India

May 2023

Navy Children School, Class X

Percentage : 97%

Visakhapatnam, Andhra Pradesh, India

May 2021

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon

Jun 2026 - Current

- Cut **PII** redaction latency by **92%** via **AWQ**-quantized fine-tuning of Gemma 4 on **SageMaker**, reaching **96% F1**.
- Shipped a document-sensitivity guardrail for the internal AI gateway, monitoring **15k+** uploads/month via **ELK**.
- **Business Impact:** Cut manual audit effort by **40%** and redaction costs by **85%**.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati

Dec 2025 - Jan 2026

- Optimized Llava 1.5 Pipeline serving **50+** clients by migrating inference to TensorRT alongside Triton inference server.
- **Business Impact:** Achieved GPU utilization increase by **30%** and serving **8x** clients with no additional resources.

AI Intern, Critical AI Pvt Ltd, Bengaluru

Jul 2025 - Sep 2025

- Deployed multimodal RAG pipelines (FAISS, llama-cpp, OSM, TTS, STT, MongoDB) behind FastAPI microservices.
- **Business Impact:** Added **3+** features and transitioned to on-prem LLM stack, reducing inference cost by **30%**.

AI Intern, WESEE, Ministry of Defence, New Delhi

May 2025 - Jul 2025

- Integrated MCP agents for SQL-based analytics and fine-tuned YOLOv8 pipelines (**94%** mAP@50, **68%** mAP50-95).
- **Business Impact:** Cut SME report prep time by **~90%** leveraging **prompt engineering**.

PROJECTS

Open-Source: Domino Speculative Decoding for vLLM Speculators

- Implemented Domino support for DFlash to the **85k-star** vLLM ecosystem, reporting a **18%** throughput gain on Qwen3.
- Improved acceptance length by **27%** over DFlash across 7 benchmarks, including GSM8K and HumanEval.
- Technology Stack: PyTorch, Transformers

AutoLinks: Intelligently InLinking Draft Articles for SEO

- Developed a semantic internal linking API to optimize site-wide link equity distribution via NER and vector search.
- Utilized LLM-as-a-judge, achieving **85+%** acceptance, achieving **p95** Latency of **2.1s**, lowering Gini coefficient by **20%**.
- Technology Stack: FastAPI, GLiNER, trafilatura, Qdrant

Graph-RAG Text-to-SQL Agent for Enterprise Databases

- Engineered Text-to-SQL agent using LangGraph, tested over **43** tables and **62** relationships in a **1GB** warehouse.
- Boosted multi-join query Execution Accuracy by **33%** by orchestrating a graph-supervised, RAG-indexed pipeline.
- Technology Stack: LangChain, LangGraph, Gemini, NetworkX

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript
AI & ML	PyTorch, Transformers, LangChain, LangGraph, Optuna, vLLM, LiteLLM
DevOps & MLOps	FastAPI, Kubernetes, Jenkins, Docker, MLflow, Triton, TensorRT, LangSmith
Cloud	GCP, AWS, Azure

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner:** Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- **LeetCode Knight:** Achieved a **1,925 peak contest rating** and solved **400+ DSA problems**.
- **Dean's list: Monsoon'23**
- **Semifinalist speaker at SMVIT Parliamentary Debate'24**